

KARTHI SRINIVASAN

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LinkedIn ◊ Personal Website

SUMMARY

Ph.D. in Computer Engineering with strong background in mathematics, computer science and machine learning. Driven by opportunity to combine technical expertise and creativity to solve open-ended problems in constantly evolving environments. Eager to apply skills to quantitative trading strategies.

EDUCATION

Yale University

Jul 2022 - Jun 2026

Ph.D. in Electrical and Computer Engineering

Advisor: Prof. Rajit Manohar

Relevant Courses: Stochastic Processes, Neural Networks and Learning Systems, Reinforcement Learning and Game Theory, Computer Architecture, Asynchronous Circuit Design

Indian Institute of Technology Madras

Jul 2017 - Jun 2022

B.Tech+M.Tech in Electrical Engineering

Overall GPA: 9.23/10.00

Relevant Courses: Applied Linear Algebra, Convex Optimization, Advanced Probability Theory, Nonlinear Control Systems, Operations Research, Information Theory, Complex Analysis

RELEVANT EXPERIENCE

BNP Paribas CIB

Jun 2025 - Aug 2025

Quantitative Research and Trading - Internship

New York, NY

- Rotations: **credit electronic-trading**, and prime services (pricing and analytics).
- Developed **automated CLO portfolio optimization system**, resulting in **1% par** improvement, with no cash usage and parity across other metrics.
- Implemented system for **identifying risk-reducing trades**, directly resulting in **trades worth 3% of trading book value** over 2 weeks.
- Created **price prediction mechanism for LQD** (bond ETF), using TRACE and fair value data.
- Analyzed convertible arbitrage hedge fund portfolios to optimize pricing of security financing.

TECHNICAL STRENGTHS

Programming Languages Libraries

Python, kdb+/q, C/C++, Haskell, MATLAB, R, LaTeX, Bash
Pandas, OR-Tools, Statsmodels, Scipy, CVXPY, Scikit-learn,
Tensorflow, SHAP, C++ STL, Boost, Google Benchmark

PUBLICATIONS

- Karthi Srinivasan, and Rajit Manohar. “**Automated Decomposition of Concurrent Programs for Asynchronous Logic Synthesis**” 2025 29th IEEE International Symposium on Asynchronous Circuits and Systems (ASYNC). IEEE, 2025.
- Karthi Srinivasan, and Rajit Manohar. “**Maelstrom: A Logic Synthesis Technique for Asynchronous Circuits**” IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD). IEEE, 2025
- Karthi Srinivasan, Yoram Moses, and Rajit Manohar. “**Opportunistic Mutual Exclusion**” 2023 28th IEEE International Symposium on Asynchronous Circuits and Systems (ASYNC). IEEE, 2023. (best paper)
- Karthi Srinivasan, and Glenn Cowan. “**Subthreshold CMOS Implementation of the Izhikevich Neuron Model**” 2022 IEEE International Symposium on Circuits and Systems (ISCAS). IEEE, 2022.
- Shubham Pande, Karthi Srinivasan, Suresh Balanethiram, Bhaswar Chakrabarti, and Anjan Chakravorty. “**2D-Motion Detection using SNNs with Graphene-Insulator-Graphene Memristive Synapses**” 2022 29th IEEE International Conference on Electronics, Circuits and Systems (ICECS). IEEE, 2022.

TEACHING

Yale University

Graduate Teaching Fellow

Aug 2023 - May 2025

New Haven, CT

- VLSI System Design (Fall 2025), Digital Systems (Spring 2025, Spring 2024), Silicon Compilation (Fall 2024), Introduction to Electronics (Fall 2023)

Indian Institute of Technology Madras

Teaching Assistant

Aug 2021 - Jun 2022

Chennai, India

- Analog Systems and Lab (Spring 2022), Neuromorphic Computing (Fall 2021)

TECHNICAL EXPERIENCE

Concordia University

MITACS Globalink Research - Internship

May 2021 - Jul 2021

Montreal, Canada

- Developed CMOS **state-of-the-art implementation** of the Izhikevich biological neuron model for energy-efficient spiking neural networks, in commercial technology node.
- Demonstrated improvements over state-of-the-art in terms of **energy (250%) and area (110%)**.

Texas Instruments Inc.

Analog IC Engineering - Internship

May 2020 - Jul 2020

Bangalore, India

- Developed novel ML-based testing framework for silicon wafers; trained on **multi-TB datasets** produced by silicon fabrication facilities.
- Achieved 20% reduction in testing time for automotive IC products; solution was **incorporated into global standard testing flow**.

RESEARCH PROJECTS

Ph.D. Thesis: Optimizing Compiler Framework for Asynchronous Circuits

Aug 2023 - Present

Yale University

- Developed **state-of-the-art compilation framework** for asynchronous circuits from high-level concurrent program descriptions.
- Compiler implementation involved **high-performance, parallel C++**.
- Demonstrated **improvements (up to 4×)** over prior work for compilation methodologies across all metrics.
- Fabricated custom chip entirely synthesized by the compiler, with over 100M transistors.
- Implemented **optimization techniques for automatic pipelining** of synthesized circuits, resulting in further performance **improvements (up to 2.3×)**.

Master's Thesis: Unsupervised Learning in Spiking Neural Networks

Sep 2021 - Jun 2022

Indian Institute of Technology Madras

- Explored use of lateral inhibition in shallow spiking neural networks to create self-organizing maps.
- Designed SNN models for low-power classification of tactile datasets.

Quasiconvex Relaxations for l_0 Optimization Problems

Jan 2020 - May 2020

Indian Institute of Technology Madras

REPORT

- Proposed **new algorithm to convert l_0 optimization problems** to more tractable quasiconvex forms.
- Implemented algorithm in MATLAB for an image compression task.
- Demonstrated superior performance compared to conventional l_1 relaxation on some tasks.

Modular Object Tracking Gimbal

May 2018 - Dec 2018

Indian Institute of Technology Madras

GITHUB

- Developed and fabricated 3-axis object-tracking gimbal system with particle-filter-based tracking and cubic time regression.
- Developed tracking algorithm, mobile app and motor control algorithm.
- Achieved close to state-of-the-art fidelity in non-occluded and partially occluded tracking.

EXTRA-CURRICULARS

Shaastra Super-Coordinator

IIT Madras

May 2019 - Jan 2020

Chennai, India

- Managed team of 5 coordinators and hosted events for annual technical festival (2,000+ participants).